

EVAPORATIVE HEAT REJECTION SYSTEMS • WCT SERIES

Round Bottle Counterflow FRP Cooling Towers

Aerodynamic Bottle-Design Counterflow Towers with Self-Rotating Alloy Sprinkler & Virgin PVC Honeycomb Fills

Engineered for High-Ambient Evaporative Cooling (CTI Standard Baseline): Designed to cool industrial process water from 42°C down to 32°C at a design wet-bulb temperature of 28°C with an approach of 4°C.

Win Equipments WCT-RL Series utilizes premium isophthalic polyester resin reinforced with multi-axial fiberglass strand mat. The self-rotating rotary aluminum sprinkler head distributes water evenly over thermoformed PVC honeycomb fill packs with zero static pressure loss.



COUNTERFLOW AERODYNAMIC

CTI STANDARD

ROTARY SPRINKLER

VIRGIN PVC FILLS

10-1500 TR

CAPACITY RANGE

3-5°C

APPROACH TO WET-BULB

< 0.005%

DRIFT LOSS

IP55

MOTOR WEATHERPROOFING

CORE ENGINEERING ADVANTAGES

Rotary Sprinkler Head

Low-friction aluminum alloy and brass rotary sprinkler head rotates automatically under incoming water pressure, eliminating static nozzle clogging.

UV-Resistant FRP Casing

Seamless heavy-gauge fiberglass reinforced plastic casing infused with UV inhibitors stands up to severe sun, wind, and corrosive industrial ambients.

Aerofoil Fan Blades

Direct-drive adjustable aerofoil cast aluminum and FRP fan blades provide high volumetric air movement at minimal electrical horsepower.

TARGET INDUSTRIAL APPLICATIONS

Plastic Injection Molding & Blow Molding

Die-Casting & Induction Furnace Coils

Water-Cooled Air Compressors

Chemical & Pharmaceutical Condensers

HVAC Central Chiller Water Plants

ENGINEERING SPECIFICATIONS MATRIX

WCT-RL Round Bottle Series Technical Data (10 to 500 TR)

Standard Ratings: 37°C in, 32°C out, 28°C wet bulb, 5°C range, 4°C approach

MODEL NO.	TONNAGE	WATER FLOW	MOTOR	FAN SPEED	FAN DIA	IN / OUT PIPE	DIMENSIONS (DIA × H)	OPERATING WT
Model Code	TR	m ³ /hr	HP	RPM	mm	Inches	mm	kg
WCT 010 RL	10 TR	6 m ³ /hr	0.5 HP	1440 RPM	450 mm	1½" / 1½"	Dia 930 × H 1690	280 kg
WCT 015 RL	15 TR	9 m ³ /hr	0.5 HP	1440 RPM	500 mm	1½" / 1½"	Dia 1030 × H 1740	340 kg
WCT 020 RL	20 TR	12 m ³ /hr	0.75 HP	1440 RPM	550 mm	2" / 2"	Dia 1130 × H 1800	410 kg
WCT 030 RL	30 TR	18 m ³ /hr	1.0 HP	1440 RPM	650 mm	2" / 2"	Dia 1330 × H 1920	550 kg
WCT 050 RL	50 TR	30 m ³ /hr	1.5 HP	960 RPM	800 mm	2½" / 2½"	Dia 1650 × H 2150	850 kg
WCT 075 RL	75 TR	45 m ³ /hr	2.0 HP	960 RPM	950 mm	3" / 3"	Dia 1950 × H 2300	1250 kg
WCT 100 RL	100 TR	60 m ³ /hr	3.0 HP	960 RPM	1100 mm	4" / 4"	Dia 2250 × H 2500	1650 kg
WCT 150 RL	150 TR	90 m ³ /hr	5.0 HP	960 RPM	1350 mm	5" / 5"	Dia 2750 × H 2800	2400 kg
WCT 200 RL	200 TR	120 m ³ /hr	7.5 HP	720 RPM	1550 mm	6" / 6"	Dia 3150 × H 3100	3200 kg
WCT 300 RL	300 TR	180 m ³ /hr	10.0 HP	720 RPM	1900 mm	8" / 8"	Dia 3850 × H 3500	4800 kg
WCT 500 RL	500 TR	300 m ³ /hr	15.0 HP	720 RPM	2400 mm	10" / 10"	Dia 4800 × H 4100	7800 kg

Low Sound Performance: WCT-RL models utilize wide-chord aerofoil blades running at low tip speed (720–960 RPM) to keep radiated noise below 68 dBA at 3 meters, meeting industrial park environmental guidelines.

HEAVY INDUSTRIAL HEAT REJECTION • WCT-SL SERIES

Square Modular Crossflow FRP Cooling Towers

Multi-Cell Expandable Towers with Open Gravity Basins & Low-Maintenance Access

Modular Multi-Cell Architecture (50 to 2000 TR): Allows side-by-side cell clustering to match any plant footprint or expansion phase. Open hot water basins operate under gravity without spray nozzles.

Win Equipments WCT-SL Series provides walk-in inspection access doors, heavy-gauge pultruded FRP framing, and hanging virgin PVC film fill packs. Routine cleaning and fill inspection can be completed without shutting down the entire plant cooling circuit.

CROSSFLOW MULTI-CELL

GRAVITY DISTRIBUTION

ZERO NOZZLE CLOG

PULTRUDED FRP FRAME



SQUARE CROSSFLOW ENGINEERING BENEFITS

Gravity Water Basin

Unpressurized top gravity basins allow maintenance technicians to clean distribution orifices during operation without turning off the circulation pump.

Walk-In Access Door

Spacious internal plenum and full-height access door provide walk-in entry for quick inspection of mechanical fans, drift eliminators, and basin sumps.

Multi-Cell Flexibility

Individual cells can be valved off during low-load winter conditions or partial production shifts to conserve massive fan electrical power.

ENGINEERING SPECIFICATIONS MATRIX

WCT-SL Square Series Technical Data (10 to 500 TR)

Standard Ratings: 37°C in, 32°C out, 28°C wet bulb, 5°C range, 4°C approach

MODEL NO.	TONNAGE	WATER FLOW	MOTOR	AIR FLOW	IN / OUT PIPE	DIMENSIONS (L×W×H)	OPERATING WT
Model Code	TR	m ³ /hr	HP	CFM	Inches	mm	kg
WCT 010 SL	10 TR	6 m ³ /hr	0.5 HP	2500 CFM	1½" / 1½"	930 × 930 × 1690	280 kg
WCT 015 SL	15 TR	9 m ³ /hr	0.5 HP	3200 CFM	1½" / 1½"	1030 × 1030 × 1740	340 kg
WCT 020 SL	20 TR	12 m ³ /hr	0.75 HP	4200 CFM	2" / 2"	1130 × 1130 × 1800	410 kg
WCT 030 SL	30 TR	18 m ³ /hr	1.0 HP	6000 CFM	2" / 2"	1330 × 1330 × 1920	550 kg
WCT 050 SL	50 TR	30 m ³ /hr	1.5 HP	9500 CFM	2½" / 2½"	1650 × 1650 × 2150	850 kg
WCT 075 SL	75 TR	45 m ³ /hr	2.0 HP	14000 CFM	3" / 3"	1950 × 1950 × 2300	1250 kg
WCT 100 SL	100 TR	60 m ³ /hr	3.0 HP	18500 CFM	4" / 4"	2250 × 2250 × 2500	1650 kg
WCT 150 SL	150 TR	90 m ³ /hr	5.0 HP	27000 CFM	5" / 5"	2750 × 2750 × 2800	2400 kg
WCT 200 SL	200 TR	120 m ³ /hr	7.5 HP	35000 CFM	6" / 6"	3150 × 3150 × 3100	3200 kg
WCT 300 SL	300 TR	180 m ³ /hr	10.0 HP	52000 CFM	8" / 8"	3850 × 3850 × 3500	4800 kg
WCT 500 SL	500 TR	300 m ³ /hr	15.0 HP	85000 CFM	10" / 10"	4800 × 4800 × 4100	7800 kg

Request Technical Submittal & Commercial Quotation

Direct factory engineering consultation for custom capacity sizing, P&ID schematics & dimensional drawings.

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